Systemadministration

Introduktion

Firmware

Agenda

Interaktion mellan hårdvara - mjukvara

1. Firmware
2. POST
3. BIOS
4. UEFI
5. Formatering
6. Partitionering
7. Bootstrapping

1. Firmware

Firmware

Mjukvara integrerad i hårdvarukomponenter

T.ex. för att styra en hårddisk

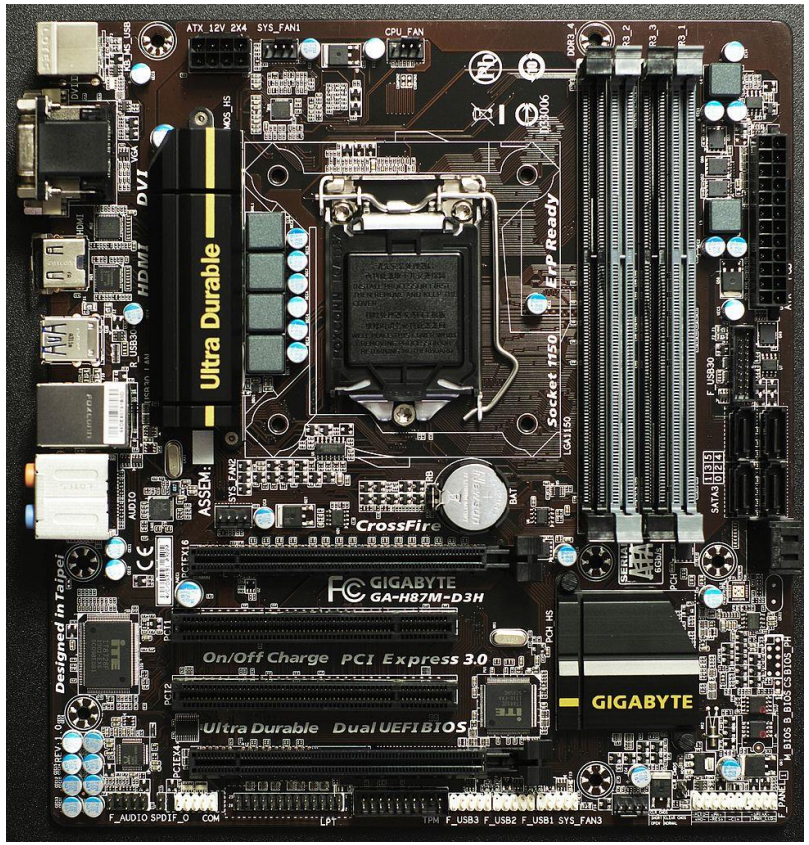
A 2.5" hard drive that has been opened, exposing its inner workings. This is a 500GB Western Digital Scorpio Blue hard drive with SATA connections. This 2.5" hard drive is common in laptops, by Evan-Amos, 2013, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Laptop-hard-drive-exposed.jpg). CC BY-SA 3.0.

Motherboard Firmware

Behöver konfigureras av användaren

Två varianter:

- **BIOS**
- **UEFI**

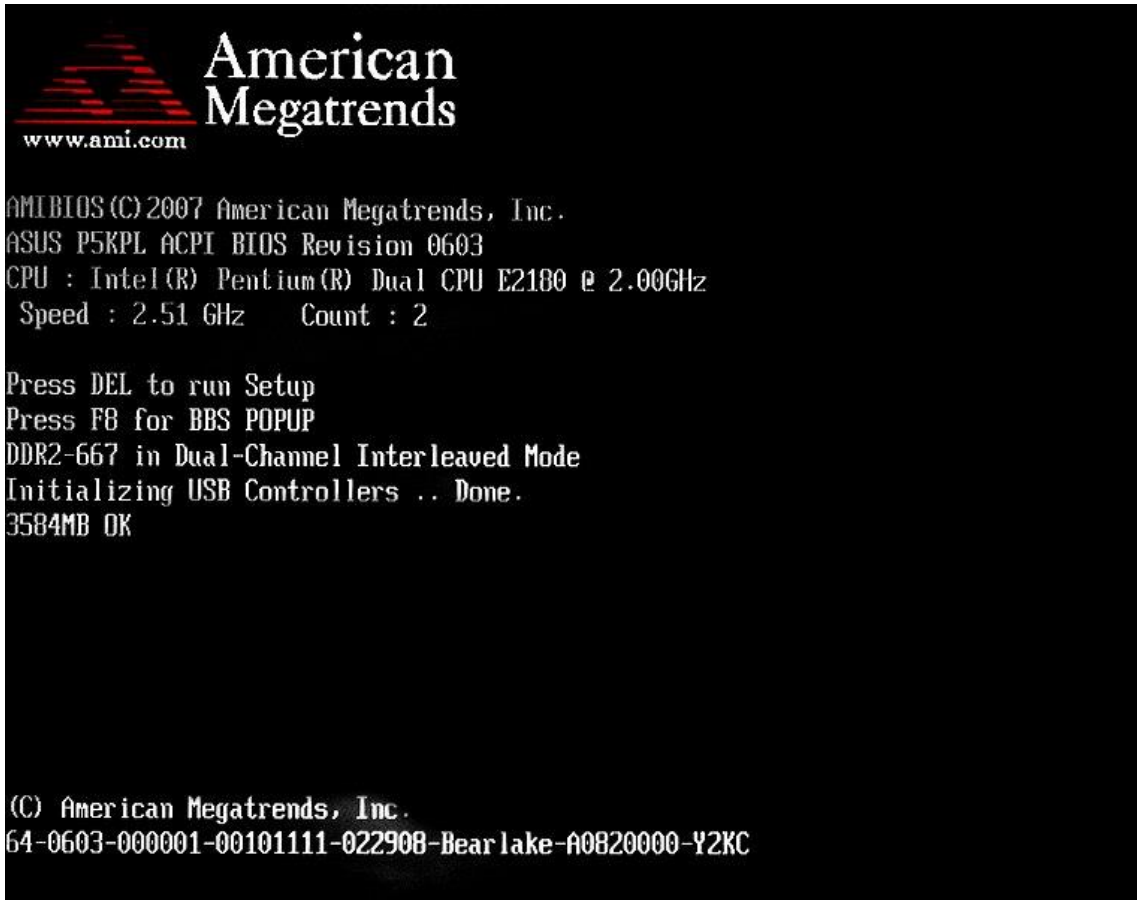


Intel core socket 1150, µATX-Mainboard Gigabyte H87M-D3H, by Smial, 2013, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Computer_motherboard_%C2%B5ATX_with_Intel_Socket_1150_IMGP8583_smial_wp.jpg). FAL 1.3.

2. POST

Power-**O**n **S**elf-**T**est

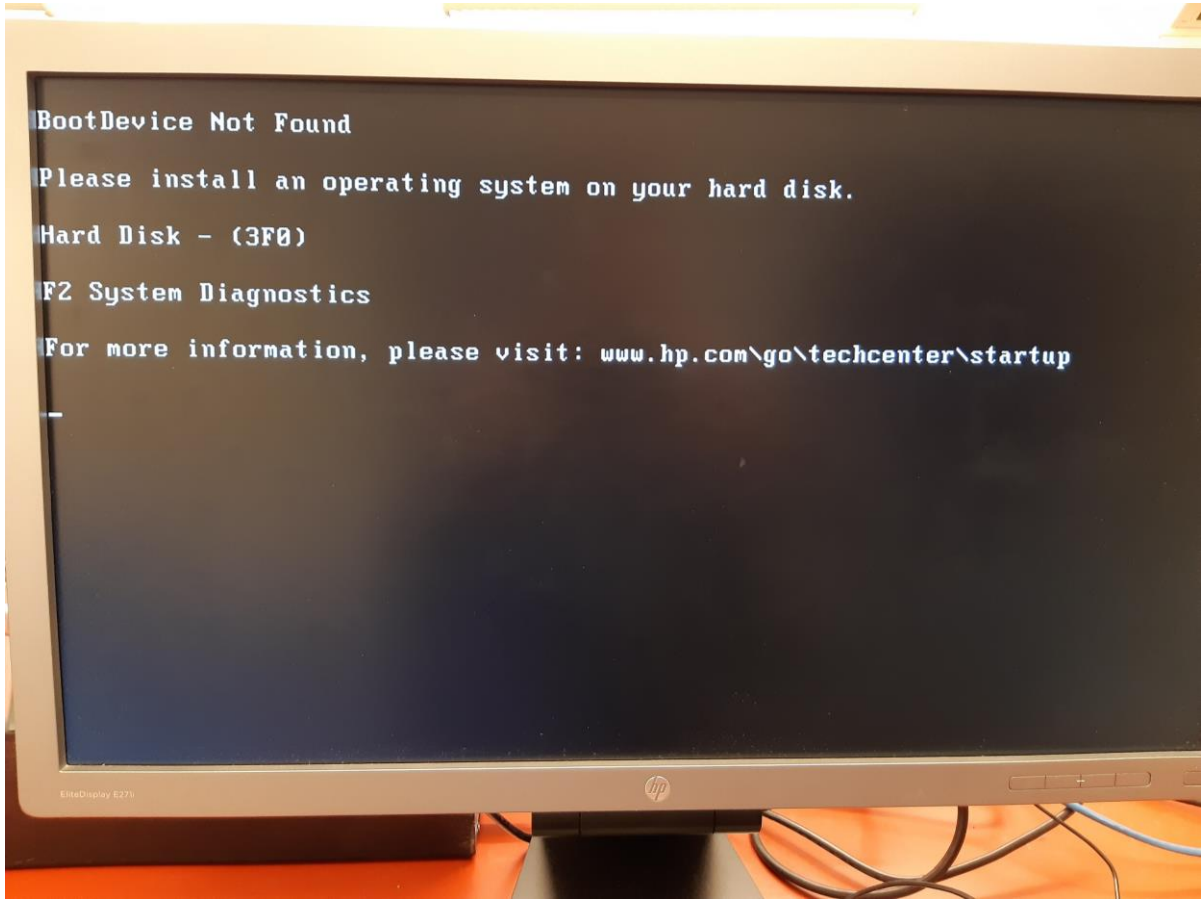
Vid uppstart körs **POST**



Power-on self test screen on a personal computer, by Macic7, 2009, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:POST_P5KPL.jpg). Public domain, trademark of American Megatrends International, LLC.

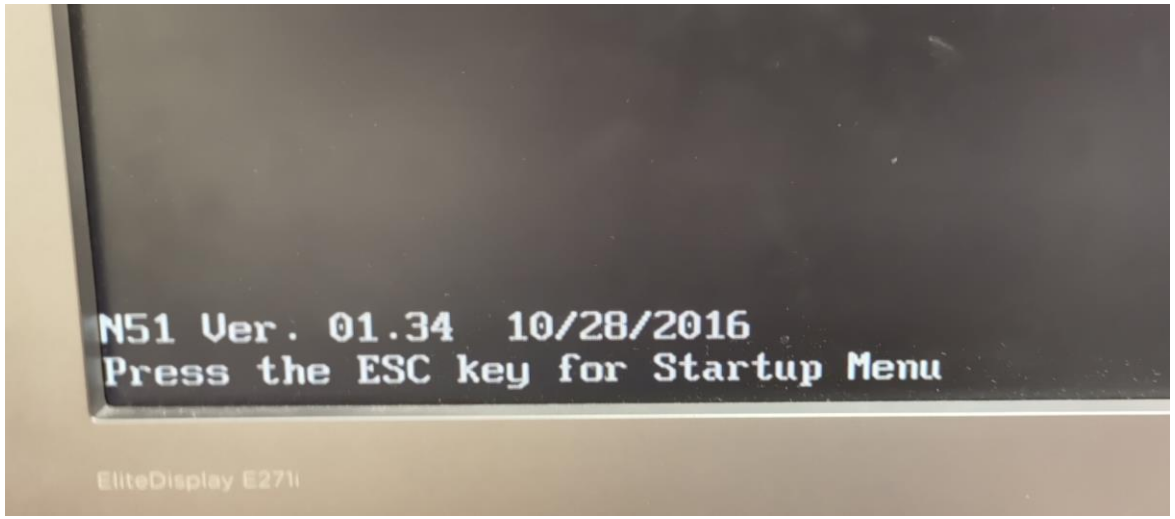
POST Error

- Error messages
- Beep codes
- Hexadecimal codes



Enter BIOS

Under **POST** tryck på **DEL, ESC, F2, F10**…



3. BIOS

Basic **I**nput/**O**utput **S**ystem

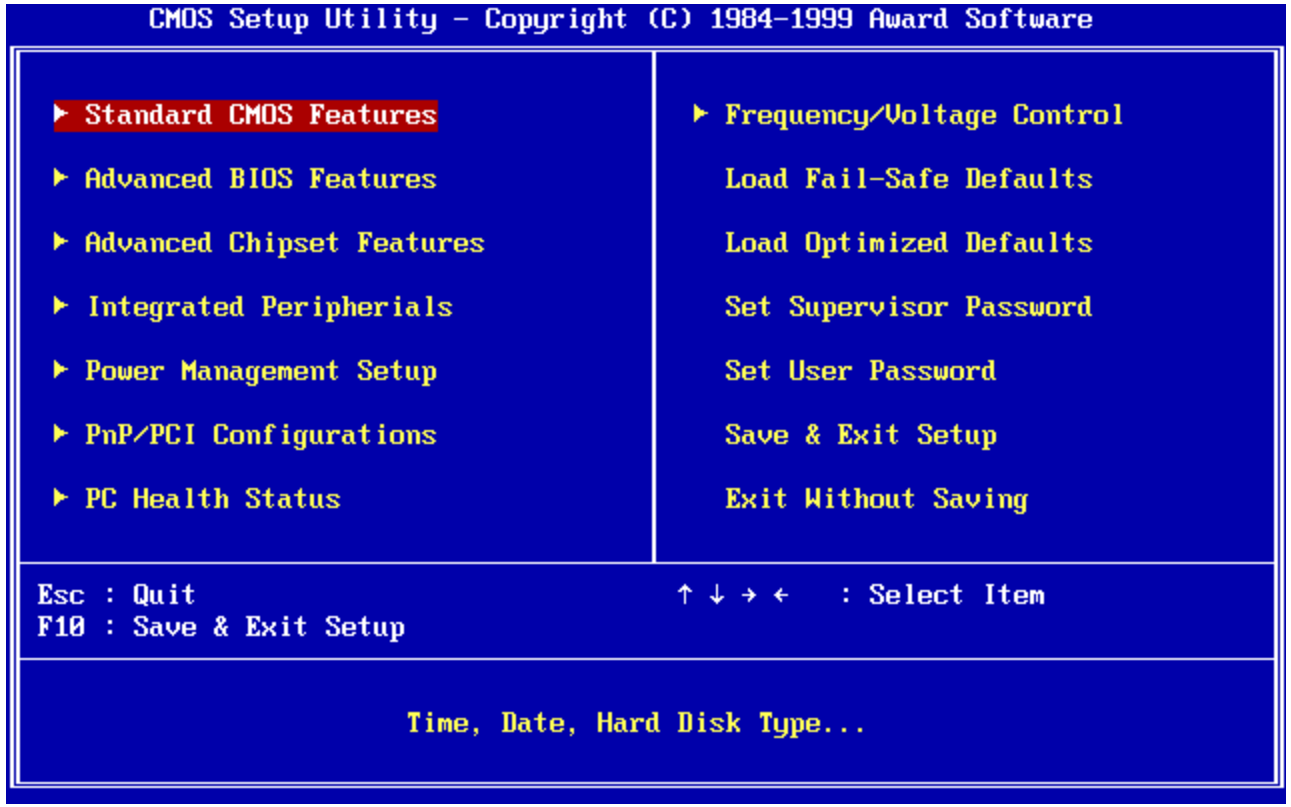
BIOS användes först 1981

Tillåter

- Kommunikation mellan CPU & andra komponenter
- Konfiguration av hårdvaruinställningar

IBM PC 5150, 1981, by Boffy b, 2022, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:IBM_PC_5150.jpg). CC BY-SA 3.0.

Basic Input/Output System



A depiction of the Award BIOS setup utility, by Kephir, 1999, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Award_BIOS_setup_utility.png). Public domain.

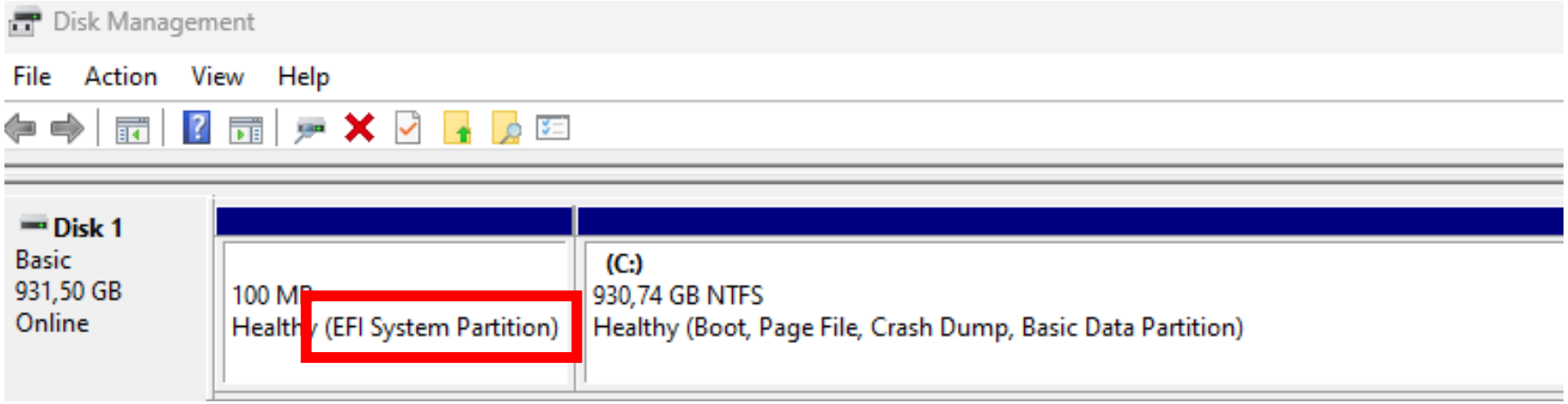
4. UEFI

BIOS limitations

- Max 2 TiB partitions
- Text-only interface
- Designed for 16-bit computers & operating systems

Extensible **F**irmware **I**nterface

1998 började Intel utveckla en ersättare: **EFI**



Unified **E**xtensible **F**irmware **I**nterface

2005 blev det en öppen standard: **UEFI**

- Inte samma **BIOS**-begränsningar
- Fler funktioner

Kallas ofta för ’**BIOS**’ ändå

UEFI logo, by UEFI Forum, 2010, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Logo_of_the_UEFI_Forum.svg). Public domain, trademark of UEFI Forum.

Asus txarte nagusien bios-a, by Beñat Arkarazo, 2019, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Bios_asus.jpg). CC BY-SA 4.0 International.

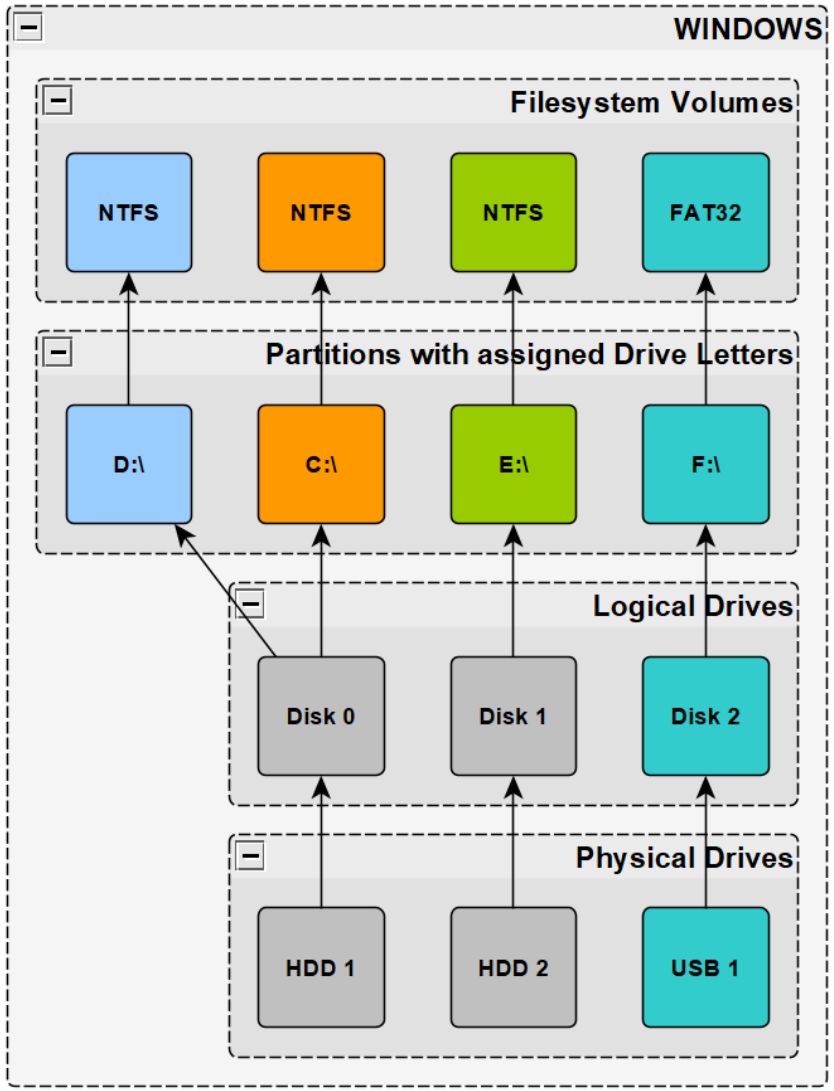
5. Formatering

Formatting

Formatering

Innan lagringsmediet kan användas

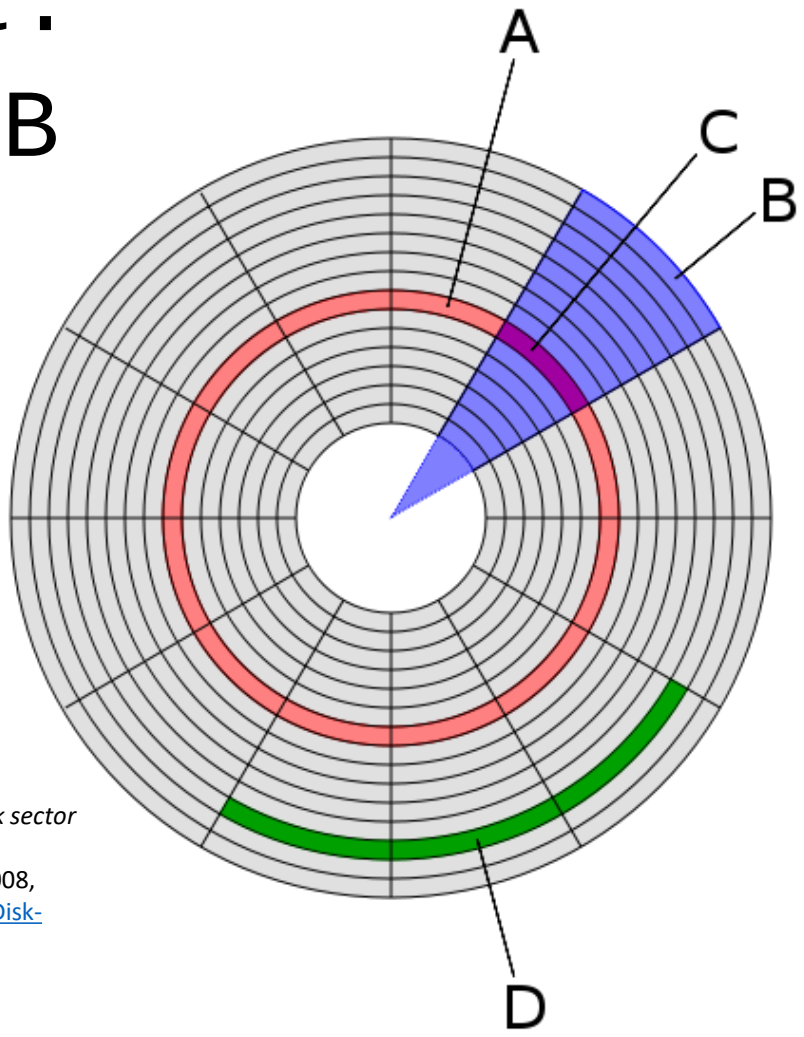
1. Low-level formatting
2. Partitioning
3. High-level formatting



Low-Level Formatting

Görs i fabrik, bland annat:

- **Sector size**, i.e. 4 KiB
- Number of **sectors**
- Number of **tracks**



Disk structure showing a track (A), a geometrical sector (B), a track sector (C) and a cluster of sectors (D). The term "sector" in the context of computer engineering usually refers to (C), by Heron2/MistWiz, 2008, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Disk-structure2.svg). Public Domain.

High-Level Formatting

Installerar ett filsystem

Volym

Volume

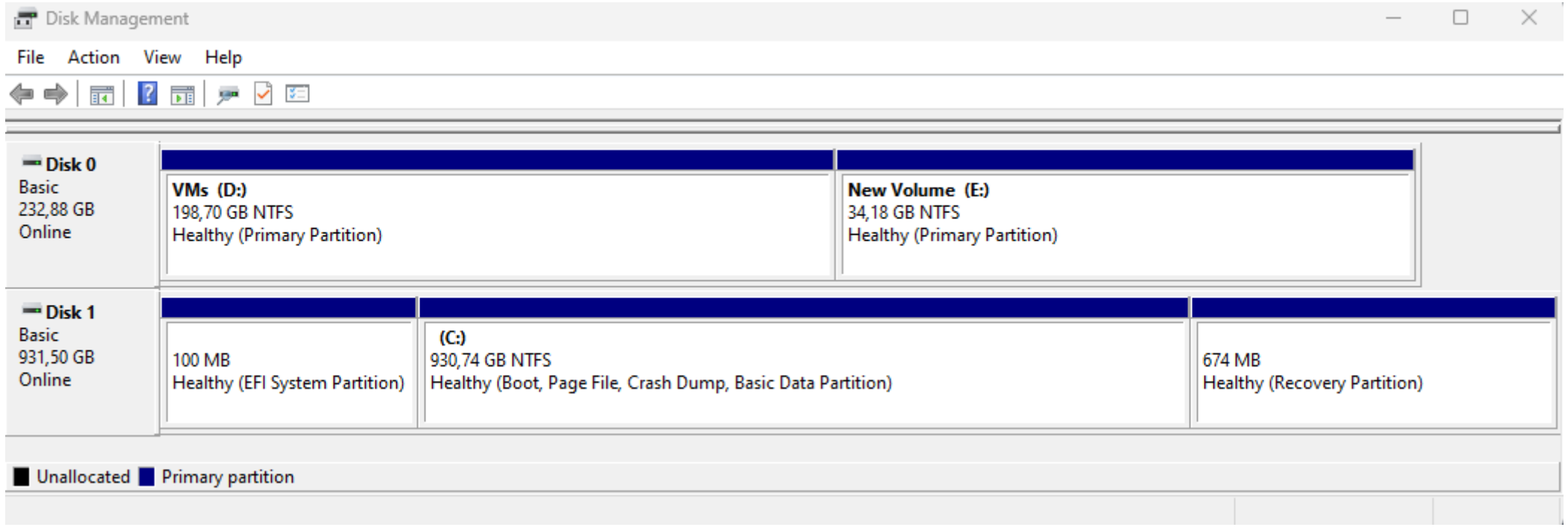
En lagringsyta med ett installerat filsystem

6. Partitionering

Partitioning

Partitionering

Delar upp ett lagringsmedium i flera logiska



Partitioning

Partitionering

Varför kan man vilja partitionera en disk?

Partitioning Types

MBR

- **BIOS**
- **UEFI CSM**, ’legacy’-mode

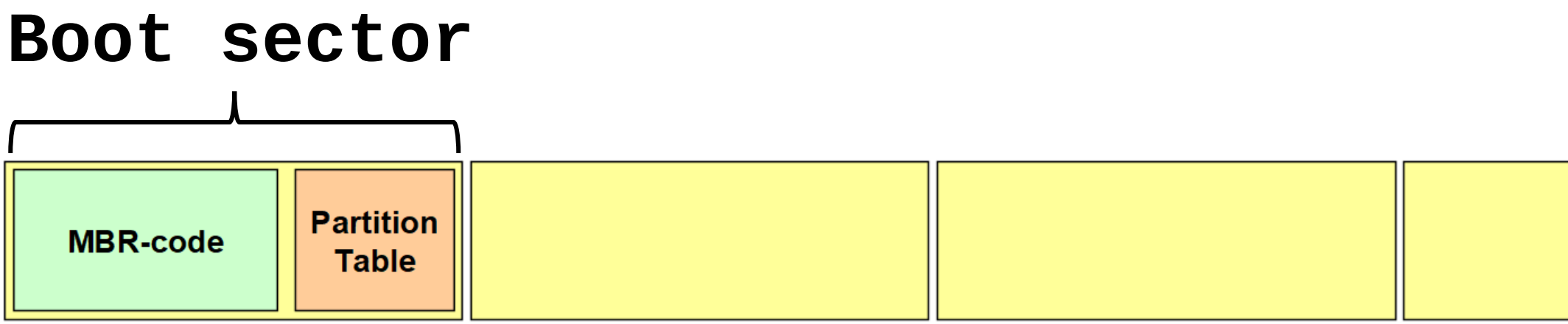
GPT

- **UEFI**

Master Boot Record

Boot sector: first sector

- MBR code (Boot loader)
- Partition table

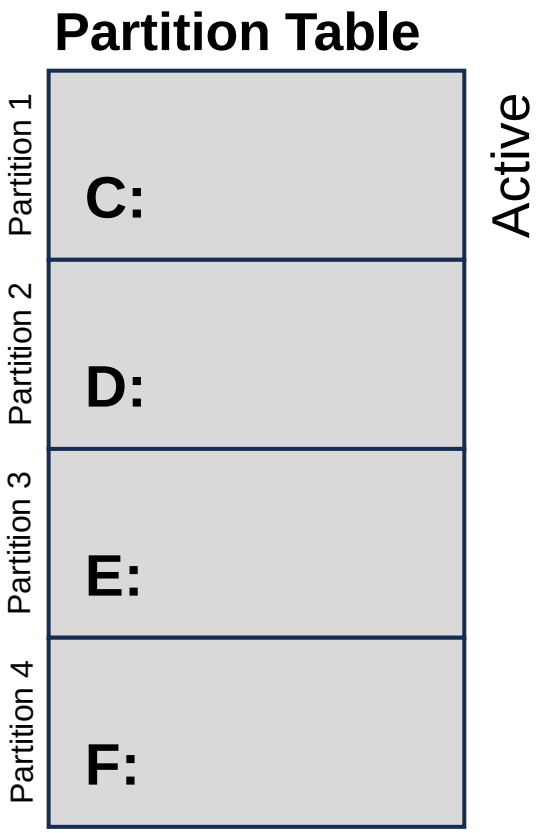


MBR Partition Table

List över alla partitioner

MBR begränsar

- Max 4 partitioner
- Max 2 TiB per partition



MBR Extended Partition

MBR begränsar

- Max 4 partitioner
- Max 2 TiB per partition

En **MBR** partition kan vara **extended**

- Max 4 **logical partitions**
- Max 2 TiB sammanlagt

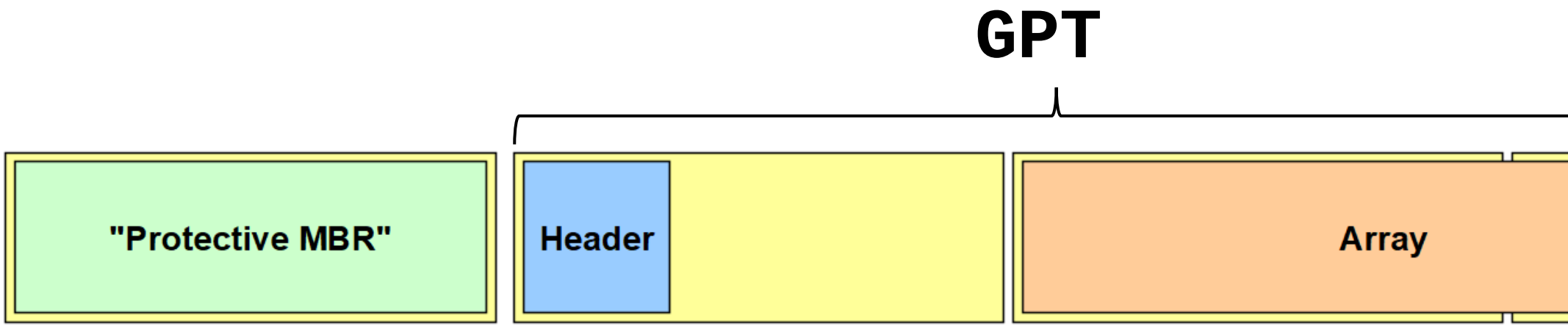
Max 7 partitioner totalt

GUID Partition Table

Inte begränsad till en sektor

Stödjer:

- Max 128 partitioner
- Max 9 ZiB stor partition

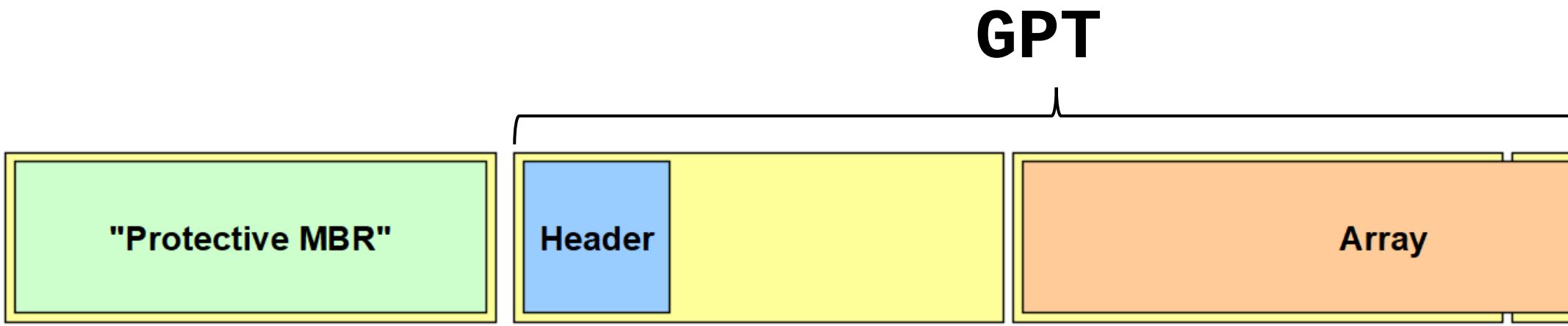


GUID Partition Table

GPT Array: lista över partitioner

GPT Header: definierar b.la. vart **GPT Array** finns

Protective MBR: En **MBR** ’varning’



GPT Fault Tolerance

GPT lagras i början och slutet av disken

GPT Header innehåller en checksumma **GPT Array**

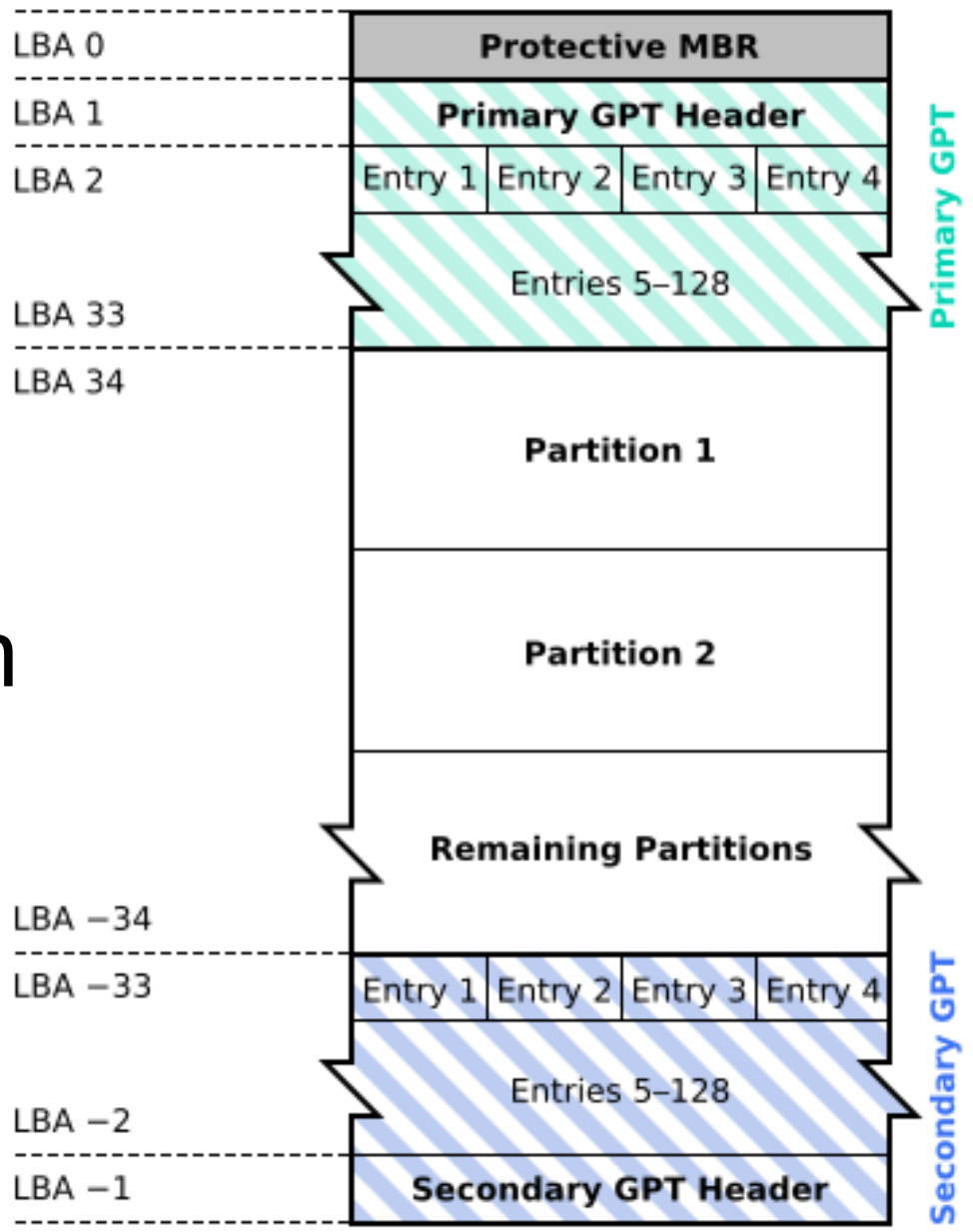
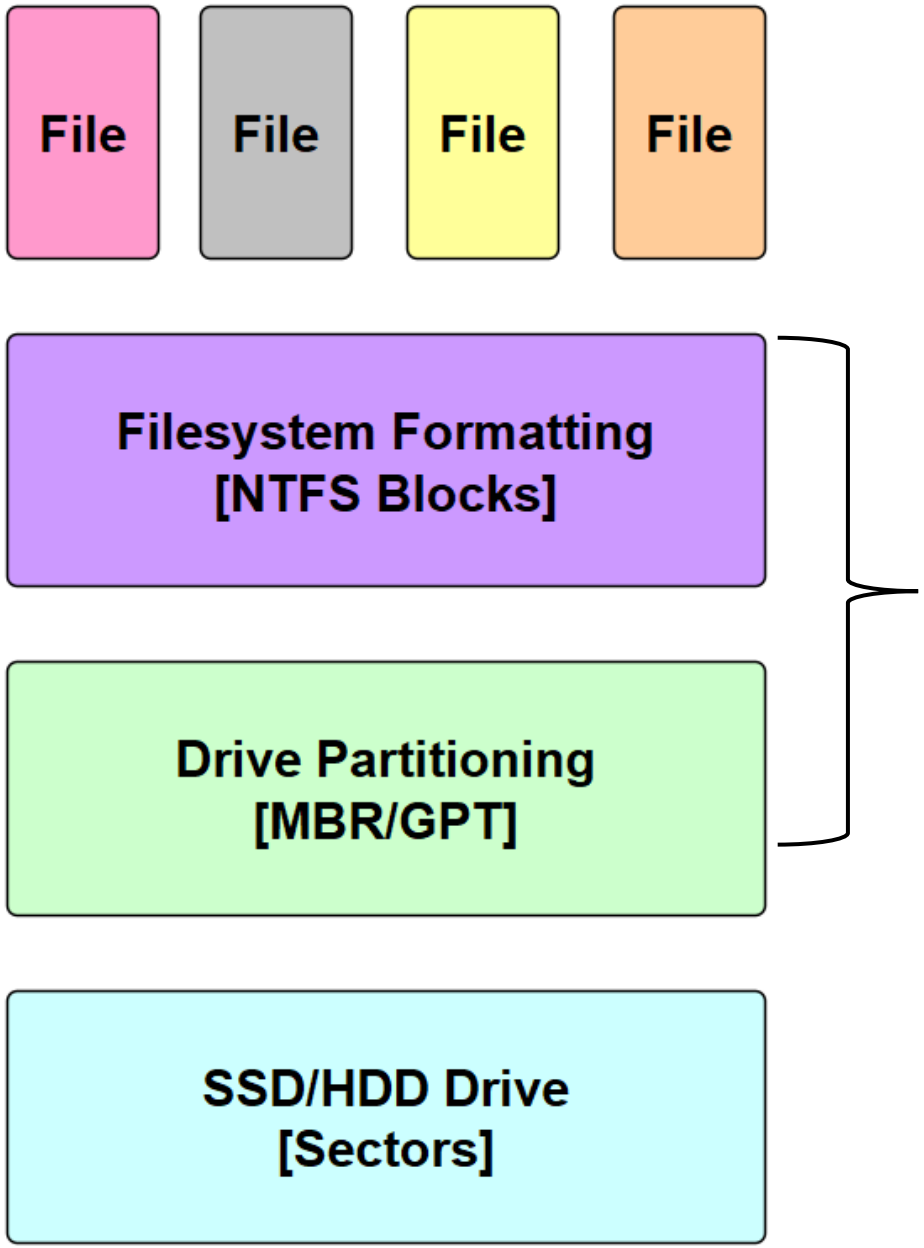
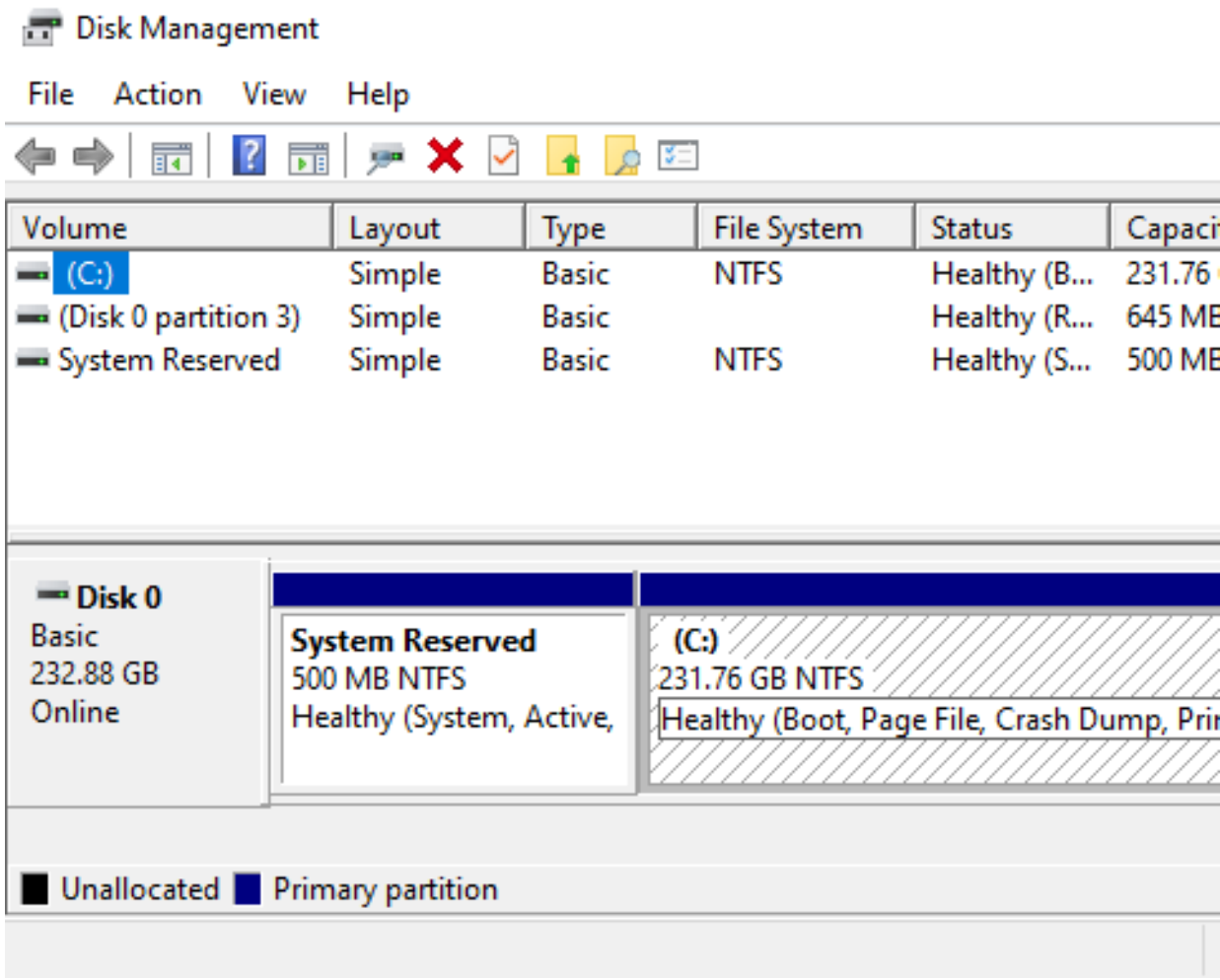


Diagram illustrating the layout of the en:GUID Partition Table (GPT) scheme. Each logical block (LBA) is 512 bytes in size. LBA addresses that are negative indicate position from the end of the volume, with −1 being the last addressable block, by Kbolino, 2007, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:GUID_Partition_Table_Scheme.svg). CC BY-SA 2.5 Generic.


File
File
File
File
Filesystem Formatting
[NTFS Blocks]
Drive Partitioning
[MBR/GPT]
SSD/HDD Drive
[Sectors]


Disk Management
File Action View Help
Volume Layout Type File System Status Capaci
(C:) Simple Basic NTFS Healthy (B... 231.76
(Disk 0 partition 3) Simple Basic Healthy (R... 645 MB
System Reserved Simple Basic NTFS Healthy (S... 500 MB
Disk 0
Basic
232.88 GB
Online
System Reserved
500 MB NTFS
Healthy (System, Active,
(C:)
231.76 GB NTFS
Healthy (Boot, Page File, Crash Dump, Pri
Unallocated
Primary partition

7. Bootstrapping

Bootstrapping

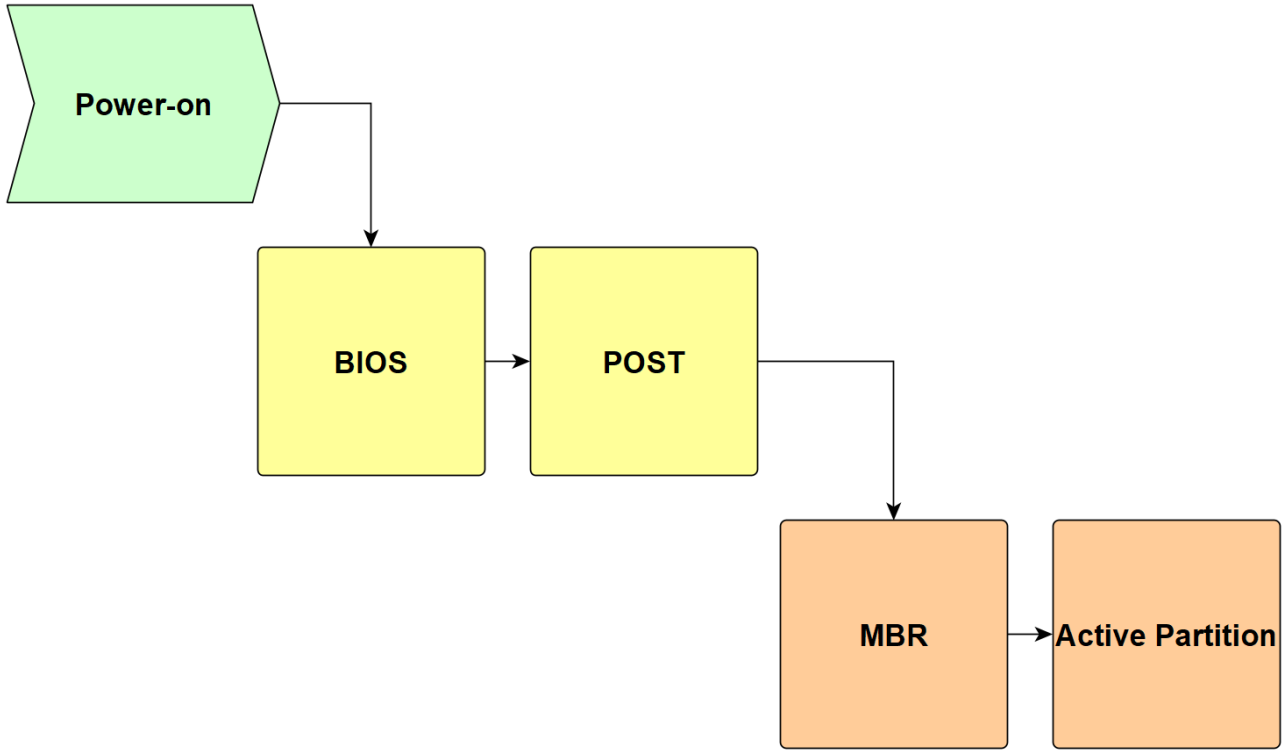
"**Booting**" is the standard term for "starting a computer". It's a shortened form of the word "**bootstrapping**", which derives from the notation that the computer has to "pull itself up by its own bootstraps"

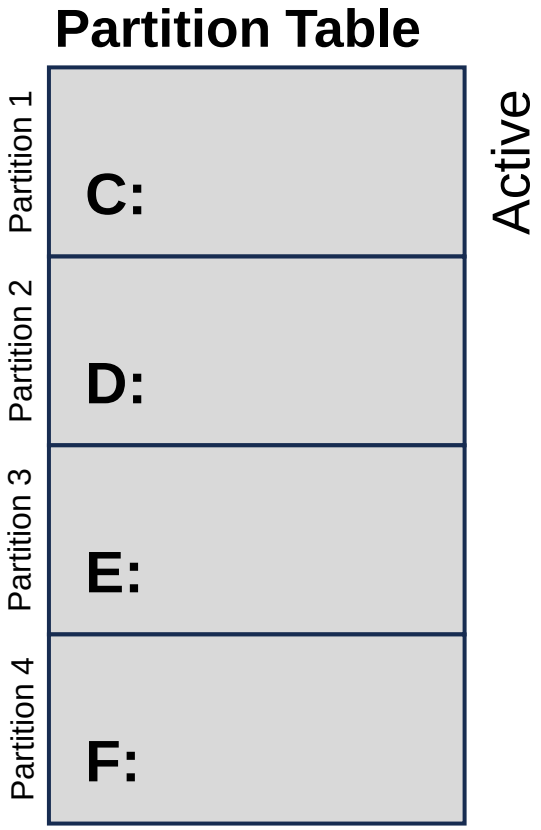
Photo of a pair of Dr. Martens boots, the displayed to show both the stitching and the sole design, by Tarquin, 2005, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Dr_Martens,_black,_old.jpg). CC BY-SA 3.0 Unported.

Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., Mackin, D. (2018). Booting and System Management Daemons. UNIX and Linux System Administration Handbook *5th ed.* (pp. 30). Pearson.

MBR Primary Partition*

En partition markeras som **aktiv**





*Microsoft kallar det för System Partition

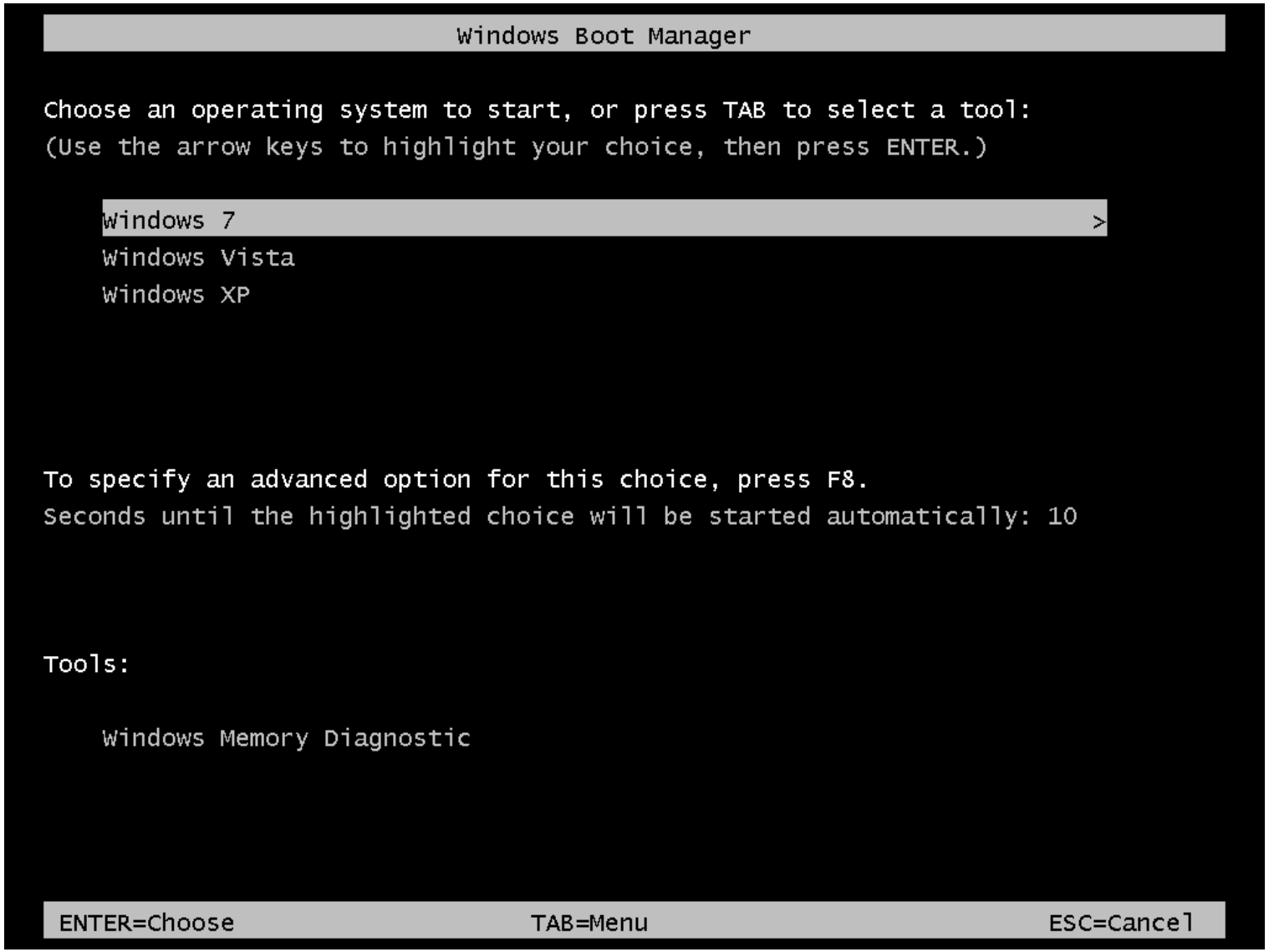
Boot Loader

BIOS/UEFI bootar vanligtvis en **boot loader**

Vilken sedan bootar OS kärnan

- Mer flexibelt: flera installerade OS
- Enklare: underlättar OS utveckling & uppdateringar

Windows Boot Manager

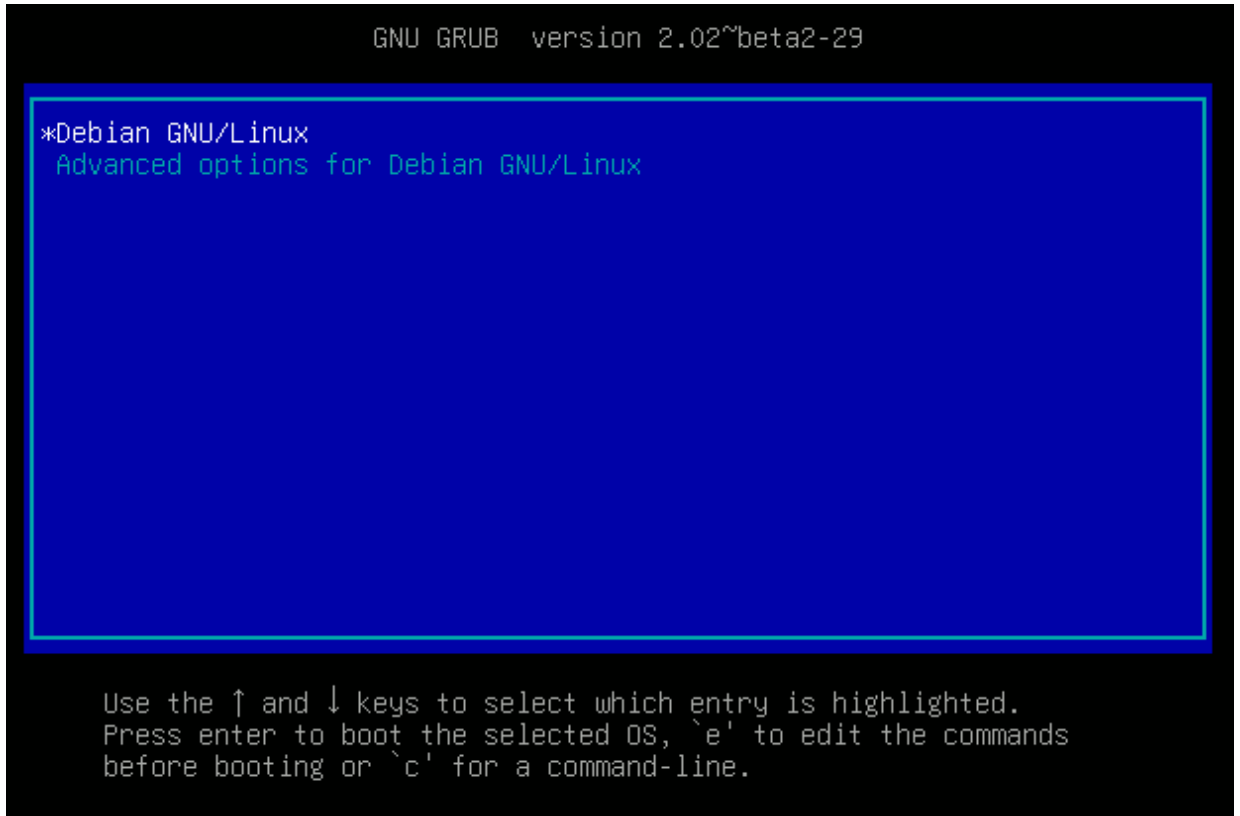


Windows Boot Manager with Windows 7 highlighted, Windows XP and Vista are also options available, by Microsoft, 2016, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Windows_Boot_Manager_with_Windows_7,Vista_and_XP.png). Public domain, trademark of Microsoft Corporation.

GNU GRUB 2.0

A larger version of the GNU GRUB logo (grub_logo.png), but without the text. The official GRUB website describes the smaller logo with text as "A GRUB logo by Karol Krenski.", by Karol Kreński, 2009, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Grub_logo_large.png). Public domain.

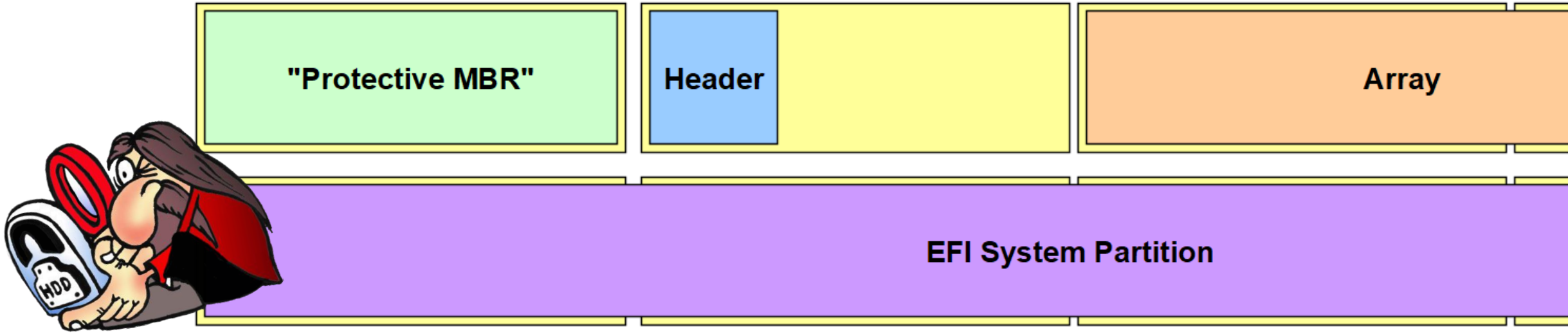
A larger version of the GNU GRUB logo (grub_logo.png), but without the text. The official GRUB website describes the smaller logo with text as "A GRUB logo by Karol Krenski.", by Karol Kreński, 2009, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Grub_logo_large.png). Public domain.



GRUB 2.02 as seen in Debian Unstable, by Huihermit, 2015, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Debian_Unstable_GRUB2_(2015).png). GPLv2.

EFI System **P**artition

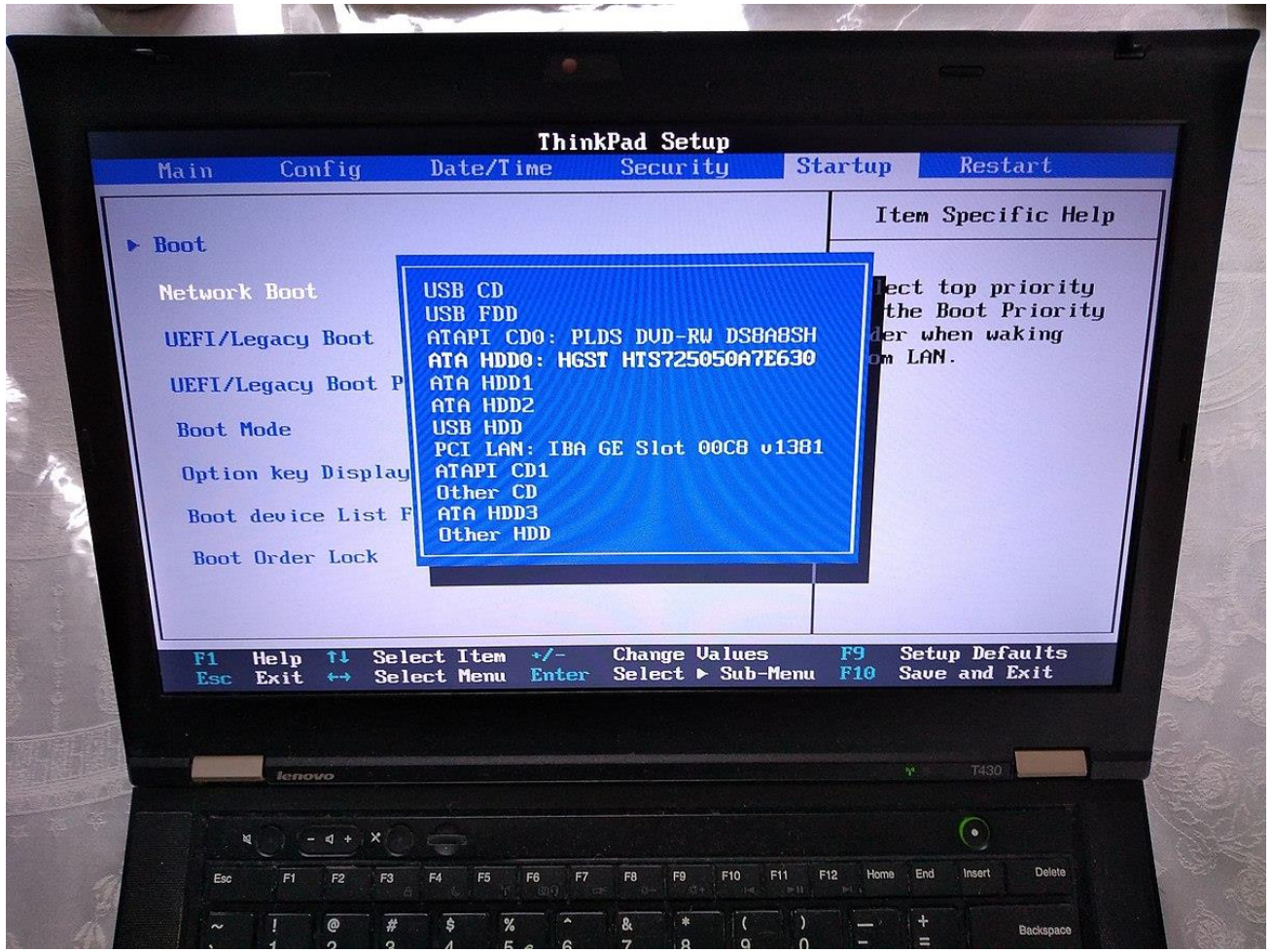
GPT boot partition kallas **ESP**



A larger version of the GNU GRUB logo (grub_logo.png), but without the text. The official GRUB website describes the smaller logo with text as "A GRUB logo by Karol Krenski.", by Karol Kreński, 2009, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Grub_logo_large.png). Public domain.

Boot Order

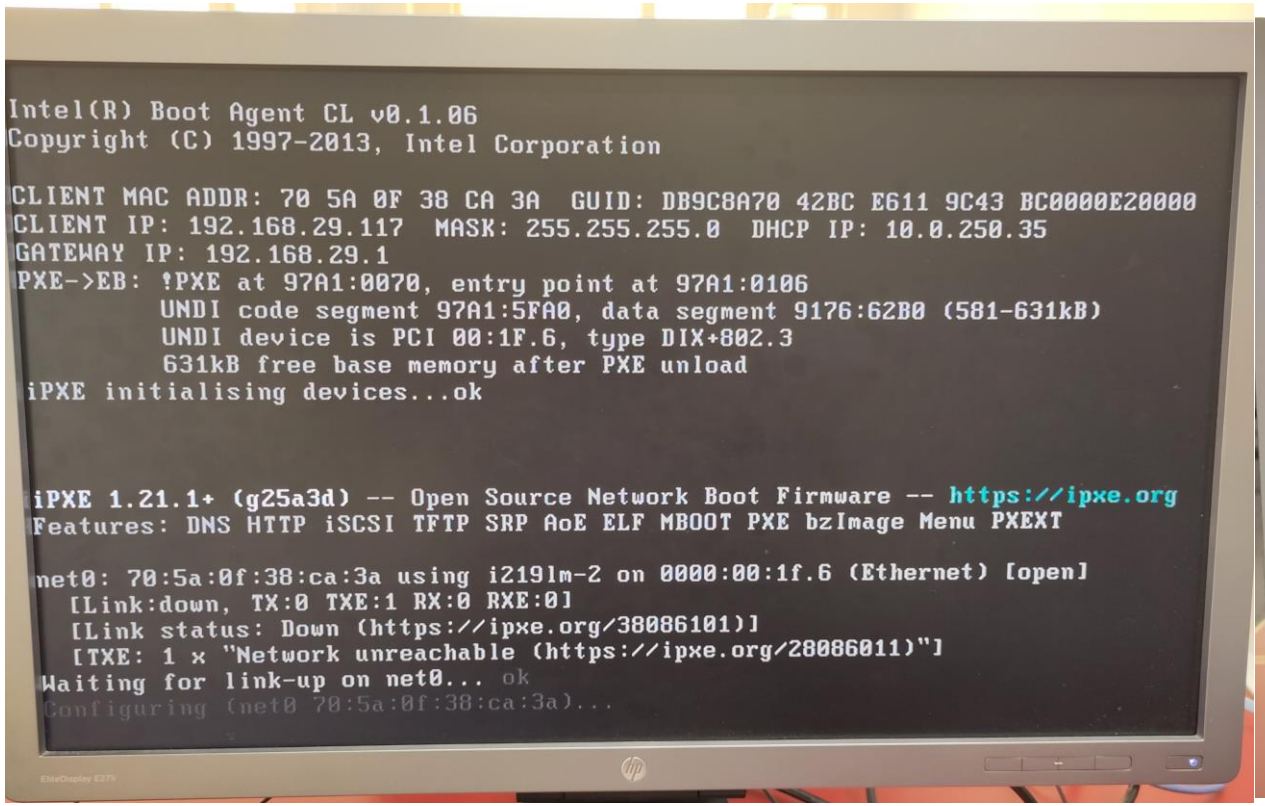
Hur vet datorn vilket sekundärminne som ska bootas?

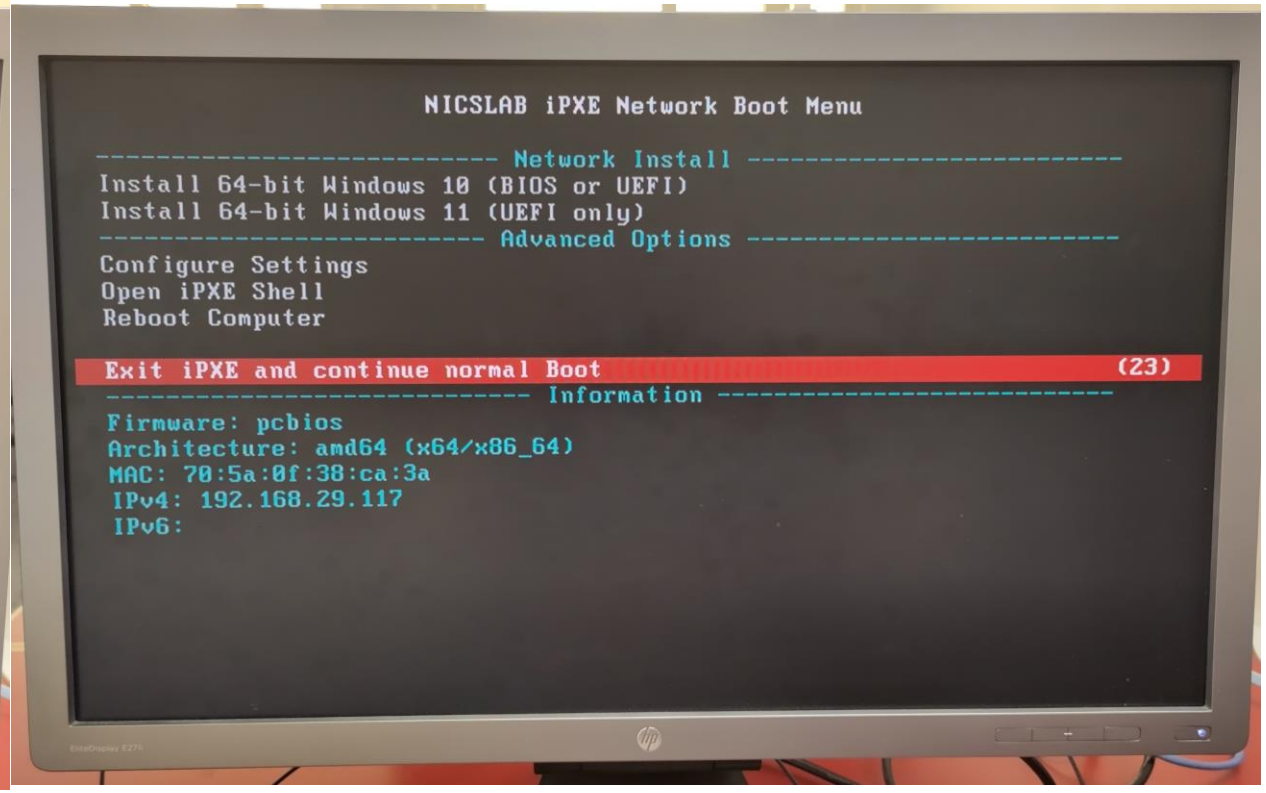


Lenovo ThinkPad T430 UEFI BIOS 1.16 boot order menu, by Cirosantilli2, 2018, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Lenovo_ThinkPad_T430_UEFI_BIOS_1.16_boot_order_menu.jpg). CC BY-SA 4.0 International.

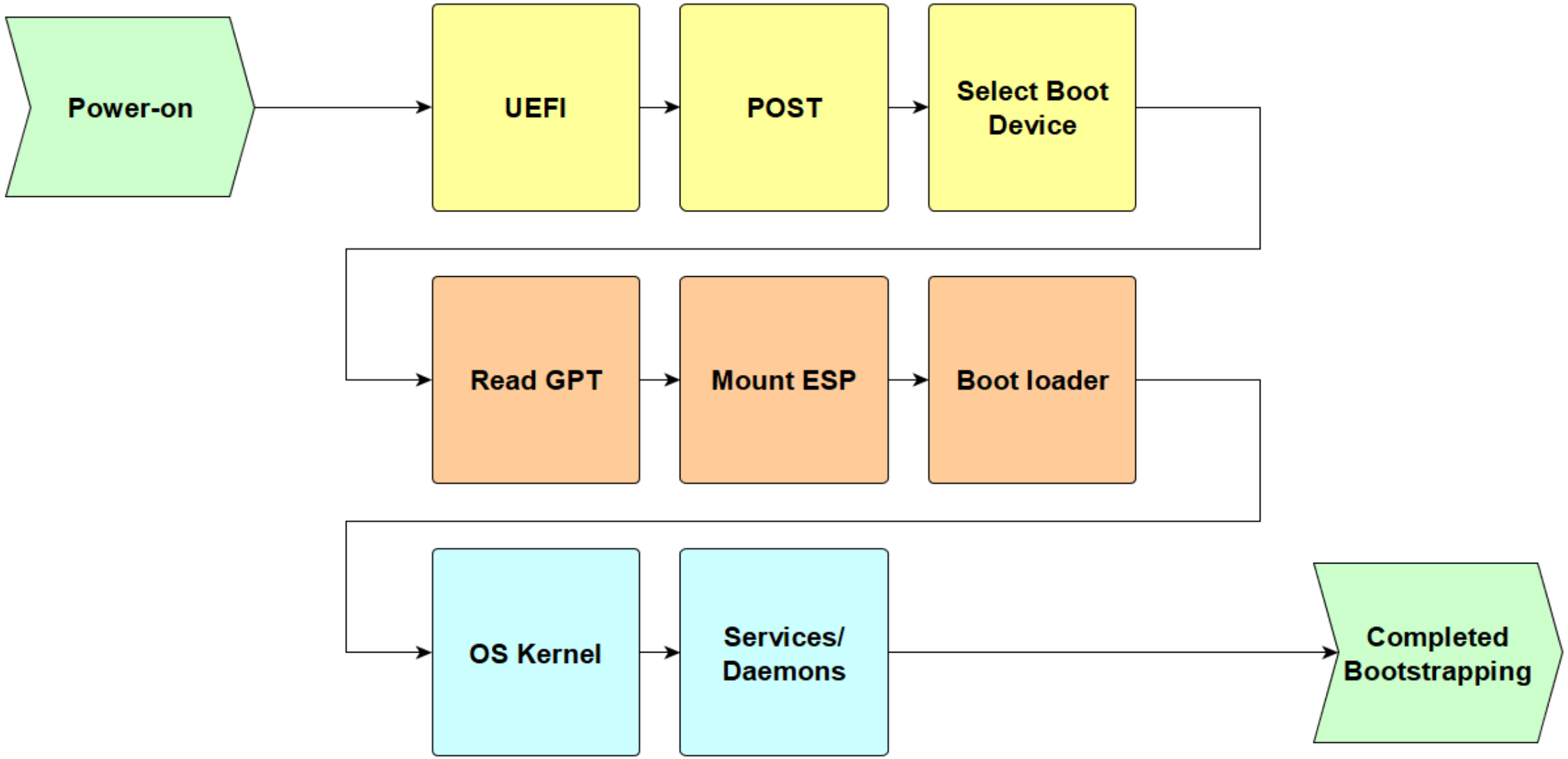
Network Boot

Använder **PXE** för att starta en boot loader över ett nätverk





Booting


Power-on
UEFI
POST
Select Boot Device
Read GPT
Mount ESP
Boot loader
OS Kernel
Services/ Daemons
Completed Bootstrapping

Till nästa gång

- Läs kurslitteraturen
- Onsdag: föreläsning
- **Fredag: ingen laboration!**

Tack för idag